
LIBERATA - GRAPH SCIENTOMETRICS FOR A SHARE BASED SYSTEM OF ACADEMIC PUBLISHING

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Abstract

Contemporary scientometric indicators remain anchored in paradigms and axioms from when academic research was done in small scholarly communities. With the proliferation of professional scientific research around the globe, academic research is now mostly conducted in large communities, with high rates of information incompleteness about work impact and individual contributions. This has profound implications for how research output is measured and quality controlled, especially as the rate of academic publishing continues to climb. Exploits of complex systems are usually found at discrete transition points, where rules turn on or off, and academia is not immune to this natural pattern. Exploitative career boosting strategies are a well recognized and growing problem within academia, largely enabled by maligned incentive structures and traditional metrics forcing discretization of credit to authors (positions) and prior works (citations) when those are fundamentally continuous valued quantities.

This article introduces Liberata's scientometrics, a share-based framework for academic publishing and quality control. In this system, authorship positions are replaced with contribution shares, which sum to unity and give both ordinality and distances of contributions. These shares can be traded on Liberata's academic marketplaces for quality control services such as peer review and replication, rewarding quality controllers based on the long term success of the work. Citations are weighted to protect against frivolous citations and credit printing, and modular correction factors are introduced to allow multiple distinct measures of impact. Liberata's metrics are formalized through two fundamental graphs: 1.) Shares and 2.) References. From these two, the Liberata system constructs the notion of *academic capital* and catalogs a naturally arising set of scientometrics that captures impact, risk, collaboration, collusion, value of quality control, diversification, etc. These metrics richly but succinctly represent academic contributions, not just for individuals, but are easily extensible to institutions, geographic regions, time periods, and research fields.

Keywords Graph Theory · Scientometrics · Econometrics · Academic Publishing · Metascience

Author's note: This article is long because it is written to be more a catalog than a single narrative. Readers are encouraged to be selective in which sections they read after section 1 and section 2.

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1 Introduction

The measurement of scholarly impact is central to the evaluation and governance of science, influencing decisions related to hiring, funding allocation, and institutional ranking. Bibliometric indicators such as publication counts, citation counts, and composite indices including the h-index (Hirsch, 2005) remain widely used to quantify research performance. A substantial body of work has examined the properties and limitations of these indicators, including numerous variants designed to improve their sensitivity to citation distributions (Bornmann, Mutz, & Daniel, 2008). Despite their prevalence, these indicators rely on simplified assumptions about the allocation of credit that are increasingly misaligned with the collaborative nature of modern scientific production.

A key limitation of existing systems lies in their treatment of authorship and citations. Most bibliometric indicators implicitly assume that publications and citations are indivisible units of credit that accrue equally to all listed coauthors. This assumption disregards the heterogeneity of contributions within collaborative research and obscures the internal structure of knowledge production. As research teams grow in size and complexity, these limitations become more pronounced.

Existing approaches have attempted to address this issue through alternative counting methods. Full counting assigns complete credit to each coauthor, leading to inflationary effects in multi-authored publications. Fractional counting mitigates this by dividing credit equally among authors, ensuring conservation of total credit, but retains the assumption of equal contribution (Waltman & van Eck, 2015). Author-order-based weighting schemes provide a partial correction by assigning credit based on position, yet these conventions vary widely across disciplines and are often inconsistent or ambiguous (Waltman, 2012). As a result, these methods remain imperfect proxies for actual contribution.

More recent work has explored network-based approaches to modeling scholarly impact. Citation network methods, including PageRank-style algorithms, use the structure of citation graphs to estimate influence within the scientific literature (Chen, Xie, Maslov, & Redner, 2007). Similarly, credit allocation models based on coauthorship and citation structures have been proposed to infer the distribution of credit among collaborators (Shen & Barabási, 2014). While these approaches incorporate relational information, they typically infer contribution indirectly rather than representing it explicitly.

In parallel, initiatives such as the CRediT taxonomy have improved transparency in authorship by providing standardized descriptions of contributor roles (Allen et al., 2014). However, these frameworks are primarily descriptive and are not directly integrated into quantitative impact measures or citation-based indicators.

Taken together, these limitations point to a fundamental gap in current scientometric systems: the absence of a unified framework that simultaneously represents heterogeneous contributions and propagates impact through the scholarly network. Addressing this gap requires moving beyond discrete authorship conventions toward representations that encode contribution structure at a finer granularity.

In this paper, we introduce *Liberata*, a share-based framework for modeling scholarly production and citation flows. In this system, contributors to a manuscript are assigned continuous contribution shares that sum to unity, replacing discrete authorship ordering. These shares define a weighted representation of the scholarly ecosystem in which citations propagate as impact signals through a network of manuscripts and contributors. The interaction between contribution shares and citation flows yields a contribution-weighted measure of impact, termed *academic capital*.

Building on this framework, we develop a set of scientometric indicators that capture not only aggregate impact but also its distribution across contributors and roles. These include measures of contribution concentration, the valuation of peer review and replication activities, and the dynamics of impact accumulation within the system. By integrating contribution shares with network-based citation analysis, the proposed approach provides a coherent and extensible foundation for contribution-aware scientometrics.

The remainder of the paper formalizes the *Liberata* framework, introduces the underlying graph representations, and develops the associated metrics for analyzing scholarly impact within this share-based system.

2 Liberata System of Open Access Publishing

In Latin, "liber" means "book" or "free", and "rata" means "rate" or "ratio". The two together can be interpreted to either mean free [market] rate, or publication shares. The fundamental idea of the Liberata open access publishing system is to use shares to denote paper contributions and to allow the trading of these shares for quality control services (peer review & replication). With the share-based system demonstrated in fig. 1, the aspiration is to liberate academia from the perverse game theory incentives and subsequent politics and rent seeking problems of the traditional academic publishing system and allow for a self-sufficient system of accurate work accreditation, incentivized quality control, open access publishing system.

2.1 Shares Based Contribution Attribution

2.1.1 Traditional Contribution Attribution

In traditional research manuscripts, work is recognized through the author list, of which there are some common conventions to interpret the positions of the authors.

- **Descending order:** In most fields of academic research, the authors are listed in descending order of perceived contribution to the work with the exception that the last author position is reserved for the principal investigator or supervisor of the project Shamoo and Resnik [2009], Sauermann and Haeussler [2017].
- **Co-first author:** this convention is used in fields where many different types of key contributions are often made to a work, such as microbiology. This practice attempts to allow multiple people to get recognition on academic metrics that count only or differently first authors from other authors Riesenbergs and Lundberg [1990], Tschardt et al. [2007].
- **Alphabetical order:** this convention is present in economics and mathematics, whereby there is no association between author position and contribution Einav and Yariv [2006], Waltman [2012]. This practice is used in fields where it is very subjective or difficult to rank order authors' contributions. In such subjective cases, discussions about rank ordering, particularly if collaborations are in small groups and likely to be repeated for many projects, impose a substantial social cost.
- **Reverse ordering of supervisors:** in some fields, the supervisors of a project (professors) are listed in reverse order of contribution where the main supervisor is at the end Tschardt et al. [2007], Bennett and Taylor [2003], and other supervisors in ascending order of contribution prior to the main supervisor. This is a practice seen in fields where it is common to have multiple professors advising on a project, and there is a need for more than the last author position to indicate that these are supervisors. Readers, upon considered inspection, could work out who is a student/worker with relatively minor contribution versus a professor/supervisor with relatively high contribution.
- **CRedit Roles:** The CRediT taxonomy Allen et al. [2014] provides more context about the types of contributions different authors have made to the paper. However, this still does not reveal how much effort was put in for each of the contributions, and how the rest of the authors value such contributions. This disconnect opens up avenues for authorship politics still. Further the roles do not provide any mechanism to include contributions from non-authors, e.g., peer reviewers and other entities who put in valuable work in making a paper higher quality.

The above conventions for recognizing contributions is a problem within academia. The fundamental issue with the traditional system is that authorship positions are a discrete system of recognition trying to track contributions that are inherently continuous, which inevitably leads to problems of unfair or inaccurate credit assignment Tschardt et al. [2007], Sauermann and Haeussler [2017]. In general, there is no agreement between scholars given an author list how precisely credit should be distributed between the authors Shen and Barabási [2014], Hagen [2008].

2.1.2 Contribution Shares in Liberata

Liberata swaps out the discrete authorship position credit system with a continuous system of contribution shares which denote percentage contribution to a manuscript. Every person on the author list is assigned a percentage contribution to a manuscript upon submission to the Liberata platform. Subsequently, other contributors such as peer reviewers or replicators may come to hold shares on the manuscript for their services to the manuscript. Let $C_m = \{c_1, c_2, \dots, c_n\}$ be the set of n contributors on a manuscript m , and